Def. A linear operator $E: V \to V$ is called a projection if $E^2 = E$.

Lemma. Let $E: V \to V$ be a projection. Then,

1) $\operatorname{Im} E = \{x \in V \mid E(x) = x\}$

2). $V = \operatorname{Im} E \oplus \ker E$

Proof. $x \in \operatorname{Im} E \Leftrightarrow x = E(\alpha)$ for some $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$. The converse is trivial.

Given $x \in V$, $x = E(x) + (x - E(x))$. Clearly $E(x) \in \operatorname{Im} E$ and

$$E(x - E(x)) = E(x) - E^2(x) = (E - E^2)(x) = 0,$$ so $x - E(x) \in \ker E$.

And, $E(y) = y$ and $E(y) = 0$, then $y = 0$. Thus $\operatorname{Im} E \cap \ker E = \{0\}$.

Thm. Let $V = W_1 \oplus W_2$. Then, there is one and only one projection $E: V \to V$ s.t.

$$\operatorname{Im} E = W_1 \quad \text{and} \quad \ker E = W_2$$

Proof Let E be a projection. For β_1, basis for $\operatorname{Im} E$, and β_2, basis for $\ker E$,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$ where $r = \operatorname{rank} E$.

Comment: In General,

let $\{\alpha_1, \cdots, \alpha_k\}$ be a basis for $\operatorname{Im} E$ and $\{\alpha_{k+1}, \cdots, \alpha_n\}$ be a basis for $\ker E$.

Then, the invertible Matrix $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_n \end{bmatrix}$ satisfies

$$EA = \begin{bmatrix} E\alpha_1 & \cdots & E\alpha_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \cdots & \alpha_k & 0 & \cdots & 0 \end{bmatrix}$$

Therefore, $E = \begin{bmatrix} \alpha_1 & \cdots & \alpha_k & 0 & \cdots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$ is we desired.

⌈Resolution of identity.

⌈Def. Given a vector space V, a collection $\{E_i\}_{i=1}^{k}$ of linear operators on V satisfying

$$\underbrace{E_i^2 = E_i \text{ for each } 1\le i\le k}_{\text{idempotent}},\quad \underbrace{E_i\circ E_j = 0 \text{ if } i\ne j}_{\text{orthogonal}},\quad \underbrace{I = E_1+\cdots+E_k}_{\text{Complete.}}$$

is called the resolution of the identity.

In fact, the orthogonal with complete implies the idempotent, because

$$I = E_1+\cdots+E_k \Rightarrow E_i = E_iE_1+\cdots+E_iE_k \Rightarrow E_i = E_i^2.$$

Moreover, if $\{E_i\}_{i=1}^{k}$ is a resolution of the identity, then

$$V = \mathrm{Im}\,E_1 \oplus \cdots \oplus \mathrm{Im}\,E_k$$

because: given $\alpha\in V$,

$$\alpha = E_1\alpha + \cdots + E_k\alpha$$

so $V = \mathrm{Im}\,E_1 + \cdots + \mathrm{Im}\,E_k$, that is, the complete gives the sum decomposition, and, for each $1\le i\le k$, $\alpha_i\in \mathrm{Im}\,E_i$ be given. Suppose that $\alpha_1+\cdots+\alpha_k=0$, then $\alpha_i = E_i\beta_i$ for some $\beta_i\in V$, so, for each $1\le j\le k$,

$$0 = E_j\cdot 0 = E_j\alpha_1+\cdots+E_j\alpha_k = E_jE_1\beta_1+\cdots+E_jE_k\beta_k \overset{\text{orthogonal}}{=} E_jE_j\beta_j \overset{\text{idempotent}}{=} E_j\beta_j = \alpha_j.$$

Thus $\mathrm{Im}\,E_i$ are independent, so proved.

⌈Lemma. Let T be a linear operator on V, and $E_1,\cdots,E_k$ be the projections satisfying

$$E_i\circ E_j = 0 \text{ if } i\ne j \quad\text{and}\quad I = \sum_{i=1}^{k} E_i.$$

Then,

For each $\mathrm{Im}\,E_i$ is invariant under T if and only if $TE_i = E_iT$ for each $1\le i\le k$.

Proof. Suppose that $\mathrm{Im}\,E_1,\cdots,\mathrm{Im}\,E_k$ are invariant under T.

Given $\alpha\in V$, $\alpha = E_1\alpha+\cdots+E_k\alpha$ and so $T\alpha = TE_1\alpha+\cdots+TE_k\alpha$.

By the assumption, each $TE_i\alpha\in \mathrm{Im}\,E_i$, so $TE_i\alpha = E_i\beta_i$ for some $\beta_i\in V$.

Therefore,

$$E_iT\alpha = E_i(TE_1\alpha+\cdots+TE_k\alpha) = E_i(E_1\beta_1+\cdots+E_k\beta_k) = E_iE_i\beta_i = E_i\beta_i = TE_i\alpha.$$

So commute. Conversely, let $\alpha\in\mathrm{Im}\,E_i$. Then, $T\alpha = TE_i\alpha = E_iT\alpha\in\mathrm{Im}\,E_i$, so proved. ☒

「Thm. Let T be a diagonalizable linear operator on a finite-dimensional space V, with distinct eigenvalues $c_1, \cdots, c_k \in F$.

Then, there exists a resolution of identity $\{E_i\}_{i=1}^{k}$ on V satisfying

$$T = c_1E_1 + \cdots + c_kE_k \quad \text{and for each } 1 \leq i \leq k,\ \operatorname{Im} E_i = \ker(T - c_iI).$$

Proof. By the equivalent condition of diagonalizable, we obtain directly that

$$V = \ker(T - c_1I) \oplus \cdots \oplus \ker(T - c_kI)$$

Now, for each $1 \leq i \leq k$, let E_i be the projection on $\ker(T - c_iI)$ along $\bigoplus_{j \neq i} \ker(T - c_jI)$

「Polynomial evaluation of linear operation.

Let T be a diagonalizable operator on a fin. dim space V with a resolution

$$T = c_1 E_1 + \cdots + c_k E_k.$$

Then, the orthogonality gives that: for any polynomial $g(t) \in F[t]$,

$$g(T) = g(c_1)E_1 + \cdots + g(c_k)E_k.$$

So, we can use the Lagrange Polynomial,

$$P_j(t) = \prod_{i \neq j} \frac{(t - c_i)}{(c_j - c_i)}, \text{ so that } P_j(T) = E_j.$$

Thm. Let V be a fin. dim. V.S over F.
If $V=\bigoplus_{i=1}^{k} W_i$, then there exist k linear operators E_i on V s.t

$$E_i \circ E_j = \delta_{ij} \cdot E_j, \quad I=\sum_{i=1}^{k} E_i, \quad W_i = \operatorname{Im} E_i \quad (1 \le i \le k) \quad \cdots (*)$$

Conversely, if $E_1, \cdots, E_k$ are k linear operators on V which satisfy $(*)$, then

$$V=\bigoplus_{i=1}^{k} W_i$$

proof. Suppose $V=\bigoplus_{i=1}^{k} W_i$. For each $1 \le j \le k$, define $E_j : V \to V : \sum_{i=1}^{k} \alpha_i \mapsto \alpha_j$. proved.
Conversely, Suppose that $E_1, \cdots, E_k$ are linear operators on V satisfying $(*)$.
Given $\alpha \in V$, $\alpha = E_1(\alpha) + \cdots + E_k(\alpha) \in W_1 + \cdots + W_k$. And, to show uniqueness such expression, let $\alpha = \alpha_1 + \cdots + \alpha_n$, where $\alpha_i \in W_i$. And, let $\alpha_i = E_i(\beta_i)$, for each $1 \le i \le k$.
Now, for each j, $1 \le j \le k$,

$$E_j(\alpha) = \sum_{i=1}^{k} E_j(\alpha_i) = \sum_{i=1}^{k} (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

layer 2

Lemma. Let T be a linear operator on V, and $E_1, \cdots, E_k$ be the projections satisfying

$$E_i \circ E_j = 0 \text{ if } i \ne j \quad \text{and} \quad I = \sum_{i=1}^{k} E_i.$$

Then,

For each $\operatorname{Im} E_i$ is invariant under T if and only if $TE_i = E_i T$ for each $1 \le i \le k$.

Proof. Suppose that $\operatorname{Im} E_1, \cdots, \operatorname{Im} E_k$ are invariant under T.
Given $\alpha \in V$, $\alpha = E_1\alpha + \cdots + E_k\alpha$ and so $T\alpha = TE_1\alpha + \cdots + TE_k\alpha$.
By the assumption, each $TE_i\alpha \in \operatorname{Im} E_i$, so $TE_i\alpha = E_i\beta_i$ for some $\beta_i \in V$.
Therefore,

$$E_i T\alpha = E_i(TE_1\alpha + \cdots + TE_k\alpha) = E_i(E_1\beta_1 + \cdots + E_k\beta_k) = E_iE_i\beta_i = E_i\beta_i = TE_i\alpha.$$

So Commute. Conversely, let $\alpha \in \operatorname{Im} E_i$. Then, $T\alpha = TE_i\alpha = E_iT\alpha \in \operatorname{Im} E_i$, so proved. ☒